

CS5487 Handout

Kernel Functions

Antoni Chan
Department of Computer Science
City University of Hong Kong

Here is a quick reference for kernel functions.

Nomenclature:

- stationary kernel: $k(x, z) = k(x - z)$
- homogeneous: $k(x, z) = k(\|x - z\|)$

Kernels on $\mathbb{R}^d$:

- linear: $k(x, z) = x^T z$
 - generally, $k(x, z) = x^T A z$, where A is a positive definite matrix.
- polynomial (inhomogeneous): $k(x, z) = (x^T z + 1)^d$
 - homogeneous: $k(x, z) = (x^T z)^d$
- Gaussian: $k(x, z) = e^{-\gamma \|x - z\|^2}$, $\gamma > 0$
- exponential: $k(x, z) = e^{-\gamma \|x - z\|}$, $\gamma > 0$

Combining kernels:

- Can also combine kernels to form kernels. See the problem set for the rules.
- e.g., linear + RBF: $k(x, z) = x^T z + e^{-\gamma \|x - z\|^2}$

Kernels on histograms:

- histograms or count vectors: $x, z \in \mathbb{R}_+^d$
- correlation: $k(x, z) = x^T z = \sum_{i=1}^d x_i z_i$
- Bhattacharyya: $k(x, z) = \sum_{i=1}^d \sqrt{x_i} \sqrt{z_i}$
- χ^2 -RBF: $k(x, z) = e^{-\gamma \chi^2(x, z)}$, $\gamma > 0$
 - χ^2 distance: $\chi^2(x, z) = \sum_{i=1}^d \frac{(x_i - z_i)^2}{\frac{1}{2}(x_i + z_i)}$
- histogram intersection: $k(x, z) = \sum_{i=1}^d \min(x_i, z_i)$

Kernels on probability distributions:

- probability density functions: $p(x), q(x)$
- correlation: $k(p, q) = \int p(x)q(x)dx$
- probability product kernel: $k(p, q) = \int p(x)^s q(x)^s dx, s > 0$
- Kullback-Leibler kernel: $k(p, q) = e^{-\gamma(D(p||q)+D(q||p))}, \gamma > 0$
 - Kullback-Leibler divergence (KLD): $D(p||q) = \int p(x) \log \frac{p(x)}{q(x)} dx$
 - KLD is not a distance metric, and thus KL kernel is **not** positive definite. Lose guarantees of the kernel algorithms.
- Jensen-Shannon kernel: $k(p, q) = e^{-\gamma JS(p, q)}, \gamma > 0$
 - mixture distribution: $m(x) = \frac{1}{2}p(x) + \frac{1}{2}q(x)$.
 - Jensen-Shannon distance (JSD): $JS(p, q) = D(p||m) + D(q||m)$
 - JSD is a distance metric, so JSK is a positive definite kernel.

Kernels on sets:

- sets of vectors: $X = \{x_1, \dots, x_n\}, Z = \{z_1, \dots, z_m\}$
- intersection kernel: $k(X, Z) = 2^{|X \cap Z|} = 2^{\text{number of common elements}}$
- pair-wise distances kernel: $k(X, Z) = e^{-\gamma \sum_{i=1}^n \sum_{j=1}^m d(x_i, z_j)}, \gamma > 0, d$ is a distance function.
- pyramid match kernel: approximation to nearest neighbor distances

Kernels on strings (graphs, trees):

- substring kernel: $k(x, z) = \sum_s w_s \phi_s(x) \phi_s(z)$
 - $\phi_s(x)$ = number of times substring s appears in x
 - substring weight $w_s > 0$
- graph and tree kernels defined analogously (substring $\rightarrow$ subgraph, subtree)